

DAY-3

BIOS or POST

troubleshooting theory

1. List 5 common BIOS or POST error messages you might see when starting a PC and write what hardware or configuration issue each one usually points to.

Here are five common BIOS/POST error messages and what they usually indicate:

1. *“CMOS Checksum Error” / “CMOS Checksum Bad”*
 - *Typical cause: Dead or weak CMOS battery on the motherboard, or corrupted BIOS settings.*
 - *Hardware/configuration issue: Motherboard’s RTC/CMOS battery needs replacement; BIOS settings may need to be reconfigured.*
2. *“Keyboard Controller Failure” / “Keyboard Error or No Keyboard Present”*

- *Typical cause: Keyboard not detected, faulty keyboard, bad PS/2 port, or controller issue on the motherboard.*
- *Hardware/configuration issue: Keyboard/connector problem or motherboard keyboard controller fault.*

3. “Boot Device Not Found” / “No Bootable Device” / “Invalid Boot Device”

- *Typical cause: BIOS can’t find a bootable OS on any drive.*
- *Hardware/configuration issue: Hard drive/SSD disconnected or failed, wrong boot order in BIOS, or corrupted OS/bootloader on the drive.*

4. “CPU Fan Error” / “CPU Fan Not Detected”

- *Typical cause: CPU fan not spinning, disconnected, or plugged into the wrong header.*
- *Hardware/configuration issue: CPU cooling fan or header problem; can also be a BIOS fan-speed threshold set too high.*

5. “Memory Parity Error” / “Memory Test Fail” / “DRAM Error”

- *Typical cause: RAM module failure, bad contact, incompatible RAM, or incorrect settings (e.g., overclock).*
- *Hardware/configuration issue: Faulty or poorly seated RAM sticks, or unstable memory configuration in BIOS.*

2. Given this scenario: On boot, a PC beeps 3 times in a repeating pattern and does not display anything on the screen. Research and write what this POST beep code typically means for an AMI BIOS, and suggest two troubleshooting steps.

What 3 beeps typically mean (AMI BIOS)

- *AMI's classic beep code tables list "3 short beeps" as a memory error, often described as base memory read/write test failure or conventional memory failure during POST.*
- *Because RAM is needed very early in POST, the BIOS cannot proceed to video initialization, so you get no display, just beeps.*

Two practical troubleshooting steps

1. Reseat and simplify the RAM configuration

- *Power off and unplug the PC.*
- *Open the case, release the RAM latches, and remove all memory modules.*
- *Clean any visible dust from the slots, then install one known-good RAM stick firmly into the primary slot (often DIMM_A1).*

- *Try to boot again and listen for changes in the beep pattern or appearance of a display.*
- *If the PC starts with a single stick but fails when others are added, one of the modules or slots is faulty.*

2. Try different RAM and/or slots

- *If reseating doesn't help, test each RAM stick individually in different slots.*
- *Use a known-good compatible RAM module if available; install it alone and attempt to boot.*
- *Persistent 3-beep errors with multiple known-good modules can indicate a motherboard memory slot or memory controller issue, in which case the board may need repair or replacement.*

3. Access the BIOS setup utility on any available PC or virtual machine, take a screenshot of the main BIOS screen, and identify the motherboard manufacturer and BIOS version.

1. Enter BIOS/UEFI

- *Restart the machine.*

- *Press the BIOS key during boot (common keys: Del, F2, F10, Esc).*
- *On many VMs (e.g., VirtualBox/VMware), you may need to enable “show boot menu” or send a specific key combination.*

2. Take a screenshot

- *On a physical PC: use the camera/phone or a screenshot tool if supported in UEFI.*
- *In a VM: use the VM software’s screenshot feature (e.g., Host+R in VirtualBox, or the VM’s “Take Screenshot” menu).*

3. Identify motherboard manufacturer and BIOS version

- *On the main BIOS screen, look for fields like:*
 - *“BIOS Vendor” or “BIOS Version” (e.g., American Megatrends Inc., BIOS Version 3.20).*
 - *“Motherboard Manufacturer”, “Board Vendor”, or “Product Name” (e.g., ASUSTeK, GIGABYTE, MSI, Dell, HP).*
- *Those two lines give you the motherboard manufacturer and BIOS version.*

4. You are given a screenshot of a POST error message that says 'CMOS checksum error – Defaults loaded.' Explain what this means and describe how you would resolve this issue in a real system.

Hint:

Think about the CMOS battery and BIOS settings reset.

